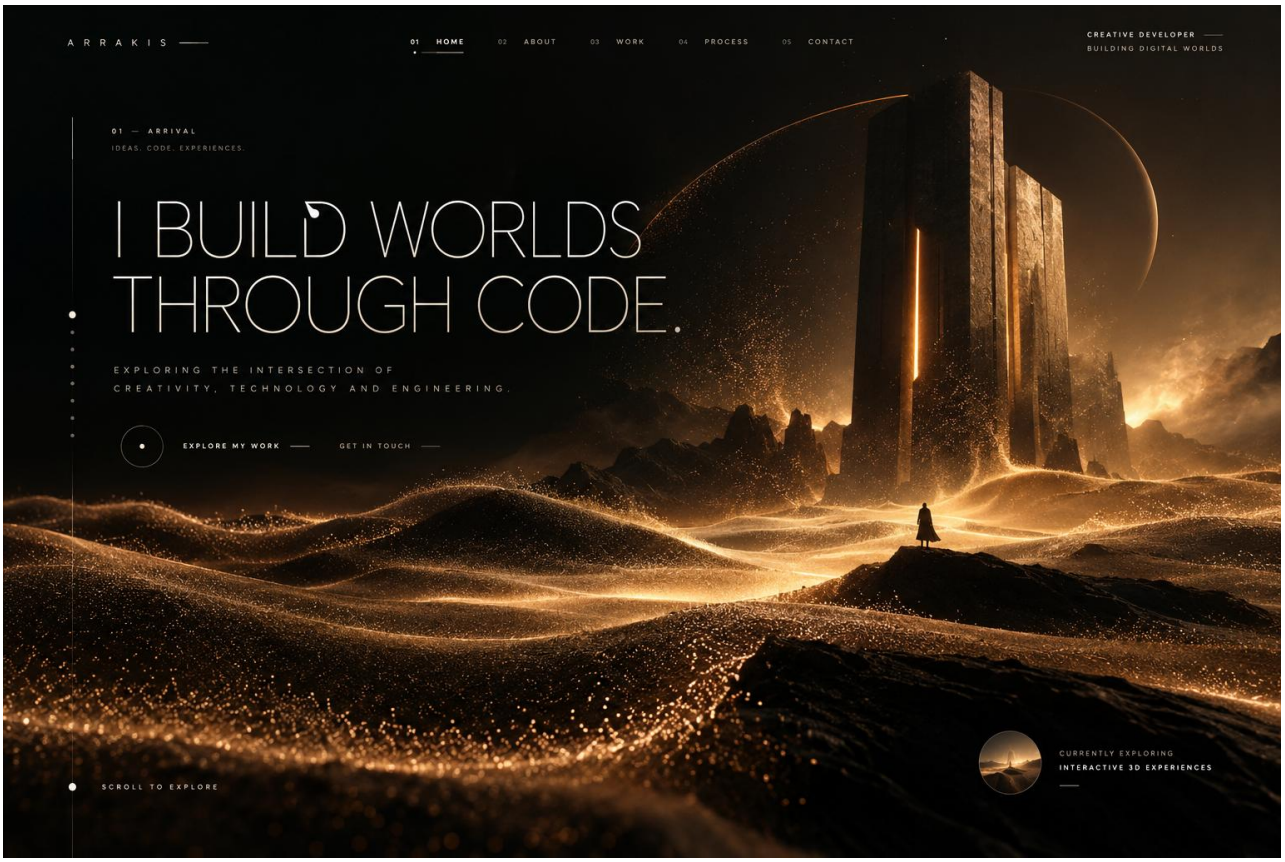
ARRAKIS

GEMINI AI STUDIO PROMPT PLAYBOOK

A complete build brief for a cinematic, scroll-driven developer portfolio



APPROVED VISUAL REFERENCE — GOLDEN PARTICLE DUNES, MONOLITH, ECLIPSE AND EDITORIAL HERO

DESIGN PRINCIPLE

The uploaded ARRAKIS image defines the visual identity. VOS9X informs the interaction language: chapter-based storytelling, visual continuity and scroll choreography.

WHAT THIS DOCUMENT CONTAINS

- A master project prompt and application architecture.
- Prompts for the homepage, About, Work, Process and Contact chapters.
- A reusable project case-study template.
- A dedicated Three.js environment, interactions and final QA prompts.

HOW TO USE THIS PLAYBOOK

Use the prompts in sequence. Do not paste the entire playbook into Gemini in one turn. Let the application build and verify each stage before moving to the next.

IMPORTANT REFERENCE RULE
Upload the approved ARRAKIS landing-page image to Gemini AI Studio before using the prompts. It is the primary source of visual truth. Use https://www.vos9x.com/ only as an interaction and storytelling reference. Do not copy its brand or artwork.

Recommended execution order

01	**Prompt 01 — Master project foundation**
02	**Prompt 02 — Complete continuous homepage**
03	**Prompt 08 — Three.js world engine**
04	**Prompts 03–06 — About, Work, Process and Contact**
05	**Prompt 07 — Individual project case-study routes**
06	**Prompt 09 — Final QA and performance audit**
07	**Prompt 10 — Hero composition refinement, if needed**

Before you start

- Open Google AI Studio and enter Build mode.
- Upload the approved hero image as a visual reference.
- Paste Prompt 01 first, then proceed in the recommended order.
- After each major stage, run the preview and fix errors before continuing.
- Keep project names and personal details editable. Do not let Gemini invent metrics, testimonials or links.
- Commit or export working versions regularly.

Project structure at a glance

- Primary experience: one continuous homepage with five navigation chapters and seven narrative beats.
- Additional routes: /work/connectville, /work/placement-portal and /work/iot-dashboard.
- 3D architecture: one persistent React Three Fiber / Three.js scene shared across homepage chapters.
- Interaction: GSAP ScrollTrigger, pointer-reactive particles, subtle camera parallax and accessible fallbacks.

PROMPT 01

Master project foundation

Establishes the design authority, narrative, technical stack, architecture and execution sequence.

COPY-READY GEMINI PROMPT

PROJECT: ARRAKIS — IMMERSIVE CREATIVE DEVELOPER PORTFOLIO

Act as a senior creative developer, digital art director, interaction designer and experienced React engineer.
Build a complete, responsive, cinematic portfolio website for Souvik Ghosh.
The approved uploaded ARRAKIS landing-page image is the PRIMARY visual design reference.
The reference website https://www.vos9x.com/ is the interaction and storytelling reference.
IMPORTANT:

- Do not copy the VOS9X brand, layout or visual identity.
- Do not turn the project into a generic dark portfolio.
- Preserve the exact golden-desert, monumental architecture and amber-light direction of the supplied ARRAKIS image.

1. CORE CREATIVE CONCEPT

Brand: ARRAKIS
Tagline: IDEAS. CODE. EXPERIENCES.
Description: Creative developer / Digital builder.
Hero headline: I BUILD WORLDS THROUGH CODE.
The website tells an original science-fiction-inspired story of discovery in a mysterious desert world.
The desert represents uncertainty and exploration. The particles represent ideas. The architecture represents engineering and structure. The artifacts represent digital products. The horizon represents future possibilities.
Use broad atmospheric themes of desert science fiction as inspiration. Do not reproduce copyrighted Dune characters, logos, scenes, dialogue, costumes or plot.

2. DESIGN REFERENCE — PRESERVE THE UPLOADED IMAGE

The reference image has a cinematic 1536 × 1024 composition.
Its main visual characteristics:

- Nearly black sky and dark negative space on the left.
- Large golden particle dunes across the foreground.
- Monumental dark sandstone monolith on the right.
- Vertical glowing amber light on the architecture.
- A subtle circular eclipse-like halo behind the structure.
- Golden atmospheric sunlight near the horizon.
- Tiny human silhouette for scale.
- Thin, oversized ivory typography.
- Minimal five-part numbered navigation.
- Vertical chapter indicator on the left edge.
- Small interactive experience preview in the lower right.

Preserve this hierarchy.
At desktop size:

- Hero headline occupies the left half.
- 3D architecture occupies the right half.
- Particle terrain fills the lower third.
- Header remains elegant and minimal.
- Text must never compete with the main architectural silhouette.

Palette:
#080909 — deep background
#F0EDE5 — primary text
#B8AA96 — secondary text
#C69A61 — desert gold
#E99A45 — amber highlights
#29231B — architectural shadow
Typography:

Use a thin geometric display typeface for the hero. Use Geist or Manrope for body and UI. Use uppercase labels with generous tracking.

3. WEBSITE STRUCTURE

Create a five-item primary navigation:
01 HOME
02 ABOUT
03 WORK
04 PROCESS
05 CONTACT
The homepage must be one continuous scroll-driven experience.
The five navigation items correspond to sections in the same cinematic environment. They are not five unrelated templates.
Additional routes:

- /work/connectville
- /work/placement-portal
- /work/iot-dashboard

Use a reusable project case-study template.

4. TECHNICAL STACK

Use:

- React
- TypeScript
- Tailwind CSS
- Three.js
- @react-three/fiber
- @react-three/drei
- GSAP
- ScrollTrigger
- @gsap/react
- Lenis
- Framer Motion

Inspect the existing project before making changes. Use the existing project structure where appropriate. Install compatible npm dependencies through the available environment. Do not replace the application framework unnecessarily.

5. ONE CONTINUOUS 3D ENVIRONMENT

Implement one persistent Three.js scene behind the complete homepage. Do NOT create seven independent 3D scenes.
The environment should consist of:

- ParticleTerrain
- SandDunes
- SandstoneMonolith
- EclipseHalo
- DesertMountains
- AtmosphericDust
- LightingSystem
- CameraRig

Keep the scene components modular.
PARTICLE DESERT:

- Thousands of GPU-rendered particles.
- Organic wave formations resembling wind-shaped desert dunes.
- Warm amber and gold particle colors.
- Depth variation and atmospheric perspective.
- Subtle turbulence.
- Cursor-responsive particle displacement.

Implement the particle field with Three.js Points and a custom ShaderMaterial where practical.
Vertex shader inputs:
uTime
uPointer
uScrollProgress
uResolution
The pointer should influence nearby particles smoothly rather than moving the whole scene abruptly.

MONOLITH:

Create a monumental architectural sandstone structure on the right. Use layered geometric meshes with slightly imperfect edges. Appearance: dark sandstone, rough surface, warm amber edge illumination, narrow vertical glowing slit and monumental scale.

Use an actual supplied GLB/GLTF asset if available, but create a working geometric fallback.

ECLIPSE AND LIGHTING:

Create a large subtle circular atmospheric object behind the monolith, with a dark central sphere or disc, thin amber ring, soft glow, warm directional sunlight and strong architectural shadows. Keep bloom restrained.

ATMOSPHERE:

Add dust particles, subtle light scattering, distant mountain silhouettes and a cinematic horizon.

6. SCROLL-DRIVEN STORYTELLING

Use GSAP ScrollTrigger. Maintain one shared scroll progress value.

Map the following chapters to their scroll ranges:

0.00–0.16 — ARRIVAL

0.16–0.30 — THE DESERT

0.30–0.44 — THE AWAKENING

0.44–0.60 — THE ARCHITECTURE

0.60–0.76 — THE ARTIFACTS

0.76–0.89 — THE METHOD

0.89–1.00 — THE HORIZON

These are narrative stages, not separate 3D applications.

During scroll:

- Move the camera gradually.
- Change particle shape and density.
- Transition lighting from dramatic golden sunrise to warmer architectural lighting.
- Reveal new architectural elements.
- Transform the monolith composition.
- Introduce project preview layers.
- Settle into the final horizon.

Avoid abrupt cuts and excessive zooming. Transitions should feel like travelling through one physical world.

7. MOUSE INTERACTION

Implement soft cursor-reactive particle displacement, subtle camera parallax, small object rotation where appropriate, interactive navigation and project previews, and restrained amber hover states.

Do not make the website difficult to read or control. Respect prefers-reduced-motion.

8. USER EXPERIENCE

Provide five-part numbered navigation, active chapter indicator, scroll progress line, smooth anchor navigation, functional project buttons, functional social links, accessible text contrast, keyboard-accessible interactions and responsive mobile navigation.

Use smooth scrolling carefully and avoid conflicts between Lenis and GSAP.

9. PERFORMANCE

- Use a single persistent 3D canvas.
- Avoid creating thousands of DOM particle elements.
- Use GPU shaders for particle motion.
- Lazy-load project screenshots.
- Reduce particle count on mobile.
- Use a static cinematic fallback on low-power devices.
- Pause or reduce animation when the tab is hidden.
- Support reduced-motion preferences.
- Prevent layout shifts.
- Ensure content remains accessible if WebGL fails.

10. CONTENT ACCURACY

Known portfolio content:

- Souvik Ghosh.
- IIT Madras BS in Data Science.
- Co-founder of Apex Circle.
- Miro Ambassador.
- Dashboard Development Intern at Flostat.

- Experience with React, TypeScript, MERN, Python and SQL.
- ConnectVille.
- Placement Portal Application.
- Real-time IoT Dashboard.

Use factual details only. Do not invent client names, revenue, awards, performance metrics or testimonials. Keep email, GitHub, LinkedIn and other external links as clearly editable placeholders until supplied.

11. DEVELOPMENT RULES

Use clean component architecture.
Suggested structure:
src/
components/
layout/
Navigation.tsx
ChapterIndicator.tsx
Footer.tsx
scene/
DesertScene.tsx
ParticleTerrain.tsx
SandstoneMonolith.tsx
Eclipse.tsx
Atmosphere.tsx
sections/
HeroSection.tsx
AboutSection.tsx
AwakeningSection.tsx
ArchitectureSection.tsx
WorkSection.tsx
ProcessSection.tsx
ContactSection.tsx
projects/
ProjectCard.tsx
CaseStudyLayout.tsx
hooks/
useScrollProgress.ts
useReducedMotion.ts
data/
profile.ts
projects.ts
pages/
HomePage.tsx
ProjectPage.tsx
styles/
Keep content data separate from components. Do not create a monolithic React component.

12. EXECUTION ORDER

Stage 1: Create application structure, global design system and navigation.
Stage 2: Build static homepage using approved reference composition.
Stage 3: Implement continuous Three.js environment.
Stage 4: Integrate cursor movement and scroll-driven camera transitions.
Stage 5: Build all content chapters and project pages.
Stage 6: Create responsive mobile layouts and performance fallbacks.
Stage 7: Test every route, interaction and animation.
Do not claim a feature is implemented unless the actual code works.
Start by creating the complete application structure and static, visually accurate homepage. Then continue with the 3D implementation without redesigning the approved hero.

PROMPT 02

Complete homepage and scroll experience

Builds the continuous seven-beat narrative while retaining five primary navigation items.

COPY-READY GEMINI PROMPT

IMPLEMENTATION 01 — THE COMPLETE ARRAKIS HOMEPAGE

Build the entire ARRAKIS homepage with the approved hero image as the visual reference.
This is the central experience of the portfolio.
Do not create separate full-page templates for Home, About, Work, Process and Contact. Implement them as chapters of one continuous scroll experience.

A. GLOBAL LAYOUT

Create a long-scrolling page containing seven narrative chapters.
Each chapter should occupy approximately one viewport or more, depending on its content.
Use a persistent fixed Three.js canvas covering the viewport behind the website content.
The HTML content should scroll normally above the canvas. The canvas should never intercept normal text selection, links, buttons or page scrolling.
The entire environment should remain visually connected.

B. HEADER

Create a minimal persistent header.
Left: ARRAKIS — thin wordmark.
Center: 01 HOME / 02 ABOUT / 03 WORK / 04 PROCESS / 05 CONTACT.
Right: CREATIVE DEVELOPER / BUILDING DIGITAL WORLDS.
Behavior:

- Clicking a navigation item scrolls to its corresponding chapter.
- Active chapter updates automatically.
- Active number gets a tiny amber marker.
- Header background becomes slightly darker when content passes behind it.
- Navigation remains visible without dominating the scene.

C. CHAPTER 01 — ARRIVAL

Section ID: home.
Use the uploaded reference composition as closely as possible.
Positioning:

- Left: headline and CTA.
- Right: gigantic sandstone monolith.
- Background: eclipse and cinematic desert horizon.
- Bottom: golden particle dunes.

Label: 01 — ARRIVAL
Headline:
I BUILD WORLDS
THROUGH CODE.
Description:
EXPLORING THE INTERSECTION OF
CREATIVITY, TECHNOLOGY AND ENGINEERING.
Buttons:
EXPLORE MY WORK
GET IN TOUCH
Create an impressive initial reveal:

1. Deep black screen.
2. A few distant particles appear.
3. Golden dunes become visible.
4. The monolith is revealed.
5. Main headline fades and moves into position.

6. Navigation and small interface details finish appearing.

Keep the entrance elegant rather than overly theatrical.

D. CHAPTER 02 — THE DESERT

Section ID: about.

Narrative: The visitor has arrived in a mysterious desert world.

The camera gradually moves closer to the dunes. The particle terrain becomes the dominant visual.

Headline:

EVERYTHING BEGINS

WITH CURIOSITY.

Introduce Souvik Ghosh, his approach to technology and his interests in software development, community and creative experiences.

Create a sophisticated editorial layout. The particle formations should slowly react to the pointer.

Use warm golden side lighting and deep cinematic shadows.

E. CHAPTER 03 — THE AWAKENING

Place this between About and Work as a storytelling chapter.

Narrative: A subtle field of luminous mineral-like particles appears inside the desert. They gather in the atmosphere and move toward a central architectural object.

Headline:

SOME IDEAS CHANGE

HOW WE SEE THE WORLD.

Represent curiosity, learning, exploration and experimentation. Do not invent a personal achievement here.

Use a subtle visual metaphor for discovery and transformation rather than any direct recreation of Dune's substance or plot.

F. CHAPTER 04 — THE ARCHITECTURE

Section ID: capabilities.

Narrative: Scattered particles begin joining together to create precise geometric structures.

Headline:

FROM CHAOS

TO STRUCTURE.

Introduce four engineering capabilities:

- Frontend engineering.
- Backend architecture.
- Creative development.
- Problem solving.

Use monumental geometric structures, each with small text annotations. Do not use generic skill progress bars.

The scene's architecture should gradually form from the particle terrain.

G. CHAPTER 05 — THE ARTIFACTS

Section ID: work.

Narrative: The structures transform into presentation surfaces that reveal digital projects.

Headline:

IDEAS MADE REAL.

Present:

1. ConnectVille.
2. Placement Portal.
3. Real-time IoT Dashboard.

Create large, carefully art-directed project previews. Use actual application screenshots where supplied. Otherwise use clearly identifiable editable screenshot placeholders.

Each preview must include project name, role, short description, technologies, view case study, live project and GitHub links when available.

Project interactions should feel integrated into the world.

H. CHAPTER 06 — THE METHOD

Section ID: process.

Narrative: The camera begins to move away from the completed architecture.

Headline:

GREAT EXPERIENCES

ARE BUILT WITH INTENTION.

Show the four stages:
01 DISCOVER
02 DESIGN
03 ENGINEER
04 REFINE
Each stage should have a clear visual metaphor and concise explanation.
Use geometric transitions and thin glowing connector lines.

I. CHAPTER 07 — THE HORIZON

Section ID: contact.
Narrative: The camera slowly pulls away from the monument. The particle dunes settle. The environment becomes calm, spacious and atmospheric.
Headline:
LET'S BUILD
WHAT DOESN'T
EXIST YET.
Add email, GitHub, LinkedIn, contact form and minimal footer.
The final frame should feel visually related to the original hero, almost as though the visitor has completed a journey and returned to the same world.

J. SCROLL CHOREOGRAPHY

Use a master GSAP timeline driven by ScrollTrigger.
Use one shared scroll progress value.
Camera choreography:

- Arrival: wide establishing composition.
- Desert: gradual forward camera movement.
- Awakening: particles rise and converge.
- Architecture: geometric forms emerge.
- Artifacts: structures reveal project content.
- Method: camera tracks horizontally or slightly upward.
- Horizon: gradual backward pull and settling.

Use smooth interpolation, not linear abrupt jumps.
Use chapter entry and exit animations with restrained opacity and transform changes.
Do not hijack scroll-wheel input. Do not force the visitor to scroll horizontally.

K. MOUSE INTERACTION

- Cursor creates a gentle disturbance in nearby particles.
- Monolith responds with subtle parallax.
- Particle flow changes direction gradually.
- Hovering on projects reveals a fine amber detail.
- Navigation responds immediately.
- Provide visible focus states for keyboard interaction.

L. MOBILE

Keep the same seven-chapter narrative.
On mobile:

- Use portrait-oriented camera composition.
- Simplify particle quantity.
- Remove expensive mouse-only effects.
- Retain scroll-driven movement.
- Ensure buttons and navigation remain usable.

M. REQUIRED TESTING

Test that there is no blank screen when WebGL fails, no animation blocking text, no section overlap, no incorrect chapter tracking, no horizontal overflow, no broken anchors and no animation running unnecessarily when the tab is hidden.
Implement the actual working interaction, not just visual mockups.

PROMPT 03

About — The Desert and Awakening

Implements the personal story and introduction as part of the continuous desert world.

COPY-READY GEMINI PROMPT

IMPLEMENTATION 02 — ABOUT / THE DESERT

Implement the About chapter within the existing ARRAKIS homepage.
Do not rebuild the navigation or the Three.js environment. Connect this chapter to the shared scroll progress.

VISUAL SCENE

Camera moves closer to the desert surface. Golden particles begin flowing like wind-carved sand. A distant monolith remains visible to provide visual continuity. The color temperature is warm amber, with strong dark shadows.

INTRODUCTION

Chapter label:
02 — THE DESERT
Headline:
EVERYTHING BEGINS
WITH CURIOSITY.
Description:
"I enjoy exploring technology, designing useful digital products and turning complex problems into thoughtful digital experiences."

PERSONAL PROFILE

Name: Souvik Ghosh.
Create an editorial profile composition.
Include:

- IIT Madras BS in Data Science.
- Software development interests.
- React, TypeScript and MERN development.
- Python and SQL.
- Community building through Apex Circle.
- Miro Ambassador experience.
- Dashboard development internship at Flostat.

Use factual, editable profile content. Show an optional portrait placeholder rather than inventing a personal image.

EXPERIENCE HIGHLIGHTS

Create a minimal animated timeline.
Apex Circle: Community building, technology sessions and hackathon organization.
Flostat: React and TypeScript dashboard development, MQTT and REST API integration.
Education: IIT Madras BS in Data Science.
Use small dates and brief descriptions where available.

AWAKENING TRANSITION

After the profile section, introduce a subtle golden particle phenomenon.
Particles rise above the desert surface and begin gathering into geometric arrangements.
Headline:
SOME IDEAS CHANGE
HOW WE SEE THE WORLD.
Supporting copy:
"Curiosity keeps me exploring. Engineering helps me turn discoveries into something useful."
This should lead directly into the Architecture chapter.

ANIMATIONS

- Slow camera approach.
- Sand particle flow.
- Gentle particle movement around the cursor.

- Staggered text reveals.
- Smooth chapter transition.

Do not create new independent canvases. Do not use generic profile cards. Keep the page visually consistent with the hero.

PROMPT 04

Work — The Artifacts

Creates the project showcase using real project content and reusable project components.

COPY-READY GEMINI PROMPT

IMPLEMENTATION 03 — WORK / THE ARTIFACTS

Implement the Work chapter in the existing ARRAKIS homepage.
Continue using the same Three.js canvas and shared scroll choreography.

CHAPTER OPENING

Chapter: 03 — THE ARTIFACTS
Headline:
IDEAS
MADE REAL.
Supporting text:
"Selected digital products, systems and interfaces built with curiosity and engineering."

SCENE TRANSFORMATION

The architectural monolith from the hero should remain part of the environment.
As the visitor reaches this chapter, parts of the architectural structure reveal layered presentation surfaces.
These surfaces should transition into actual project showcase layouts.
Do not make them resemble generic browser windows.

PROJECT 001 — CONNECTVILLE

Category: DIGITAL PRODUCT / FULL-STACK
Description: A platform for discovering and engaging with events and communities.
Technologies: Use editable project metadata.
Large screenshot placeholder.
Actions:
VIEW CASE STUDY
LIVE PROJECT
GITHUB

PROJECT 002 — PLACEMENT PORTAL

Category: WEB APPLICATION / ENGINEERING
Description: A campus recruitment platform supporting administrators, companies and students.
Showcase:

- Admin dashboard.
- Company drive management.
- Student application workflow.
- Placement analytics.

Use real project screenshots where available.

PROJECT 003 — IOT DASHBOARD

Category: REAL-TIME DATA / FRONTEND
Description: An interface for monitoring IoT water management systems.
Show:

- Sensor readings.
- Data visualization.
- Live monitoring interface.
- Responsive dashboard layout.

PRESENTATION

Use large project images with asymmetric composition.
Each project should have enough negative space. Alternate text and image positioning.
Include project number, title, short description, technology labels, live project button, GitHub button and case study button.

PROJECT INTERACTIONS

On hover:

- Image scale increases slightly.
- A subtle amber border becomes visible.
- Project metadata moves into focus.
- Cursor changes to a clear interactive indicator.

On click: navigate to the corresponding case study route.

CLOSING

Headline:
EVERY BUILD
LEAVES A TRACE.
Continue into the Process chapter.
Do not invent performance numbers or client testimonials. Use reusable project data and components.

PROMPT 05

Process — The Method

Shows the developer's approach through a transformation from scattered particles to precise architecture.

COPY-READY GEMINI PROMPT

IMPLEMENTATION 04 — PROCESS / THE METHOD

Implement the Process chapter inside the existing ARRAKIS homepage.
This should communicate how the developer thinks, plans and engineers products.
Preserve the continuous Three.js scene.

OPENING

Chapter: 04 — THE METHOD
Headline:
GREAT EXPERIENCES
ARE BUILT WITH INTENTION.
Background: The previously assembled monolithic structures.
As the visitor scrolls, the architecture begins separating into smaller organized components. This represents the process of breaking complex problems into manageable systems.

STAGE 01 — DISCOVER

Label: 01 / UNDERSTAND THE PROBLEM
Description: Understand the requirements, users, context and constraints before choosing a technical solution.
Visual: Scattered particles representing initial ideas.

STAGE 02 — DESIGN

Label: 02 / SHAPE THE EXPERIENCE
Description: Establish information architecture, interaction patterns, component structure and visual direction.
Visual: Particles form clear geometric patterns.

STAGE 03 — ENGINEER

Label: 03 / BUILD THE SYSTEM
Description: Transform the design into reusable components, connected APIs, database operations and functional interfaces.
Visual: Geometric structures become precise architectural systems.

STAGE 04 — REFINE

Label: 04 / PERFECT THE DETAILS
Description: Test responsiveness, accessibility, performance and user interactions.
Visual: The architecture becomes complete and illuminated by warm golden light.

TECHNICAL CAPABILITIES

Create a compact editorial technology section.
Frontend: React, Next.js, TypeScript, Tailwind CSS.
Backend: Node.js, Express.js, Python, Flask.
Data: MongoDB, PostgreSQL, SQL, Supabase.
Creative: Three.js, GSAP, Framer Motion.
These technologies should be editable through a central data file. Do not create percentage skill meters.

SCENE EXIT

The camera gradually moves backwards from the architectural environment. The structures align toward the distant horizon.
Headline:
PRECISION LIVES
IN THE DETAILS.
Continue smoothly into Contact.
Maintain the same typography, particles, colors and visual hierarchy as the approved hero.

PROMPT 06

Contact — The Horizon

Builds the final chapter, contact form and visual resolution of the journey.

COPY-READY GEMINI PROMPT

IMPLEMENTATION 05 — CONTACT / THE HORIZON

Implement the final chapter of the ARRAKIS homepage.
Reuse the existing 3D scene.

OPENING SCENE

Chapter: 05 — THE HORIZON
Visual: A vast desert landscape with an illuminated horizon.
The camera pulls back slowly from the monolith. Particle movement becomes calmer and more sparse. The atmosphere should feel peaceful, cinematic and open.

MAIN HEADLINE

LET'S BUILD
WHAT DOESN'T
EXIST YET.
Supporting description:
"Have an interesting idea, a product to develop or a challenging problem to solve? Let's create something meaningful."

CONTACT DETAILS

Create an elegant two-column composition.
LEFT COLUMN:
- Email address.
- GitHub profile.
- LinkedIn profile.
- Location: India.

Use editable placeholders for links that have not been supplied.
RIGHT COLUMN:
A functional contact form.
Fields:
- Name.
- Email.
- Subject.
- Message.

Button: SEND MESSAGE →

FORM BEHAVIOUR

- Validate required fields.
- Validate email format.
- Show clear submission errors.
- Show a successful submission state only after actual success.
- Use a server-side contact endpoint or a configurable email service.
- Do not put API credentials in frontend code.

FINAL BRAND MOMENT

Display the ARRAKIS wordmark.
Text: IDEAS. CODE. EXPERIENCES.
Small caption: CREATIVE DEVELOPER / DIGITAL BUILDER.
Footer navigation: HOME / ABOUT / WORK / PROCESS / CONTACT.
Include GitHub, LinkedIn and copyright.

FINAL ANIMATION

The golden particles gradually settle around the horizon. The vertical light within the distant monolith glows softly. The camera stops moving.

The final composition should visually echo the opening scene and create a feeling of closure.

IMPORTANT

No excessive animations around the form. Do not sacrifice usability for the visual experience. The final chapter must remain accessible even when WebGL is unavailable.

PROMPT 07

Reusable project case-study template

Creates detailed, data-driven project routes for ConnectVille, Placement Portal and the IoT dashboard.

COPY-READY GEMINI PROMPT

IMPLEMENTATION 06 — PROJECT CASE STUDY TEMPLATE

Create a reusable project case-study page for the ARRAKIS portfolio.
Routes:

- /work/connectville
- /work/placement-portal
- /work/iot-dashboard

Use one reusable layout that takes project information from a central data file.

DESIGN

Preserve the ARRAKIS brand identity.
The case study should have a cinematic introduction, followed by clear technical storytelling.
Maintain dark backgrounds, ivory typography, golden details and restrained desert atmosphere.
Do not repeat the full-screen Home hero on every case study.

SECTION 01 — PROJECT INTRODUCTION

Small metadata: WORK / PROJECT 001
Large dynamic heading: [PROJECT NAME]
Summary: [PROJECT SUMMARY]
Metadata: ROLE / CATEGORY / TECH STACK / TIMELINE
Add one large application screenshot.

SECTION 02 — THE CHALLENGE

Heading: THE PROBLEM.
Explain background, target users, requirements and constraints.
Use an elegant editorial text layout.

SECTION 03 — THE APPROACH

Heading: THE APPROACH.
Present the design decisions, architecture and planning.
Include research, architecture, technology selection and user experience decisions.

SECTION 04 — SYSTEM ARCHITECTURE

Heading: UNDER THE SURFACE.
Create a technically clear architecture diagram.
Possible components: Frontend, API layer, Backend, Authentication, Database and External integrations.
Only include components actually implemented by that project.

SECTION 05 — KEY FEATURES

Create a detailed visual showcase of features. Use alternating image and text layouts. Add screenshots and annotations.

SECTION 06 — ENGINEERING CHALLENGES

Heading: THE HARD PART.
Explain real implementation challenges and their solutions. Avoid generic statements.

SECTION 07 — RESULTS AND LEARNINGS

Heading: WHAT CAME TO LIFE.
Present verified results and deliverables. Do not generate imaginary statistics or testimonials.

SECTION 08 — NAVIGATION

Previous project. Next project. Back to selected work.

TECHNICAL REQUIREMENTS

- Responsive layouts.
- Dynamic project content.
- Reusable components.
- SEO metadata.
- Accessible image descriptions.
- Functional external links.

Do not create three separate page implementations. Use one reusable case study component and data-driven project records.

PROMPT 08

Three.js desert world engine

Implements the real interactive 3D environment rather than a static image or CSS approximation.

COPY-READY GEMINI PROMPT

IMPLEMENTATION 07 — ARRAKIS 3D WORLD ENGINE

Implement the complete interactive Three.js desert environment.
This is a real 3D implementation task, not a CSS illustration.

1. SCENE ARCHITECTURE

Create a persistent Three.js scene using React Three Fiber.
The scene contains:

- ParticleTerrain.
- SandDunes.
- SandstoneMonolith.
- EclipseHalo.
- DesertMountains.
- AtmosphericDust.
- LightingSystem.
- CameraRig.

Keep the scene components modular.

2. PARTICLE TERRAIN

Use THREE.Points and a custom GLSL ShaderMaterial.
Generate a dense grid of particle positions.
The particle positions should form an organic desert surface using layered noise functions and sine-based wave deformation.
Vertex shader inputs:

- uTime
- uPointer
- uScrollProgress
- uResolution

The particles should have warm golden colors, variable opacity, different particle sizes, atmospheric depth, smooth motion and cursor-reactive displacement.
The effect should resemble luminous sand dunes rather than a regular flat point grid. Avoid visible repetitive patterns.

3. SANDSTONE MONOLITH

Create a monumental vertical architectural form.
Use layered geometric meshes with slightly imperfect edges.
Appearance:

- Dark sandstone material.
- Rough surface.
- Warm amber edge illumination.
- Narrow vertical glowing slit.
- Monumental scale.

The monolith should dominate the right half of the hero composition.
Use an actual supplied GLB/GLTF asset if available, but create a working geometric fallback.

4. ECLIPSE AND LIGHTING

Create a large circular atmospheric object behind the monolith.
Use a dark central sphere or disc, thin amber ring, soft glow, warm directional sunlight and strong architectural shadows.
Keep bloom restrained.

5. MOUSE INTERACTION

Normalize pointer coordinates.
Track pointer movement and smoothly interpolate the pointer influence.
Particles should move locally around the pointer.

Do not let pointer movement cause sudden camera jumps.
Apply subtle camera parallax to the architecture.
Disable or reduce cursor effects on touch devices.

6. SCROLL INTEGRATION

Connect the scene to the master GSAP ScrollTrigger timeline.
Use one shared scroll progress value.
At each chapter:

- Update the camera position.
- Adjust the particle terrain.
- Animate monolith position and rotation subtly.
- Adjust lighting.
- Introduce new geometry when necessary.

Do not destroy or recreate the scene at each section.

7. VISUAL CONTINUITY

The opening desert must remain recognizable throughout the website.
Transitions should feel like travelling through a single environment.
Project showcase content should be a separate HTML layer, not part of the particle shader.

8. PERFORMANCE

- Use shader calculations on the GPU.
- Avoid excessive draw calls.
- Avoid unnecessary React renders on each animation frame.
- Use refs for fast-changing scene values.
- Scale particle quantity according to device capability.
- Limit device pixel ratio on mobile.
- Pause animations when hidden.
- Implement an accessible reduced-motion mode.
- Provide a cinematic static fallback when WebGL is unavailable.

9. VERIFICATION

Test actual mouse movement and scroll behaviour.
Confirm that particles really react, the camera actually changes position, each chapter triggers the intended scene state, the canvas remains continuous, the page stays scrollable, and text and controls remain interactive.
Do not mark the implementation complete until these behaviours work in the live preview.

PROMPT 09

Final QA and quality assurance

Audits visual fidelity, interactions, mobile responsiveness, performance and code quality.

COPY-READY GEMINI PROMPT

FINAL QA — ARRAKIS PORTFOLIO

Act as a senior frontend engineer and interaction designer.
Audit the entire ARRAKIS website against the uploaded approved visual reference.
Do not redesign the website.

DESIGN AUDIT

Check:

- Hero composition and image alignment.
- Headline typography.
- Navigation placement.
- Golden particle environment.
- Monolith proportions.
- Spacing and visual hierarchy.
- Consistent chapter identity.
- Project image quality.
- Footer and contact layout.

INTERACTION AUDIT

Test every navigation link, the active chapter indicator, scrolling forward and backward, mouse-reactive particles, all GSAP transitions, project hover and click interactions, and contact form validation.

RESPONSIVE AUDIT

Test:

- 1440px desktop.
- 1024px tablet.
- 768px tablet.
- 390px mobile.
- 360px small mobile.

Fix horizontal overflow, overlapping content, broken typography, excessive particle density, unusable mobile navigation, canvas blocking buttons and unexpected animation jumps.

PERFORMANCE AUDIT

Check initial page load, WebGL rendering, GPU usage, particle count, canvas pixel ratio, lazy image loading, reduced-motion behaviour and fallback when WebGL is unavailable.

CODE QUALITY

Check console errors, TypeScript errors, unused dependencies, duplicate animation timelines, event listener cleanup, correct ScrollTrigger cleanup, no unnecessary React re-renders, correct routing and reusable project components.

FINAL RULE

Preserve the approved visual design.
Fix functionality and visual inconsistencies without introducing additional sections, unrelated visual effects or unnecessary refactoring.
Report any remaining limitations honestly.

PROMPT 10

Hero composition refinement

Use when Gemini drifts from the approved reference image; this prompt restores the key composition and proportions.

COPY-READY GEMINI PROMPT

Use the attached 1536 × 1024 ARRAKIS image as the exact hero composition reference.
Match these proportions:

- Header: approximately the top 7% of the screen.
- Left progress rail: approximately 5% from the left edge.
- Chapter label: around 8.5% from the left and 14% from the top.
- Main headline: around 8.5% from the left, occupying the left 47% of the screen.
- Monolith: positioned in the right 30% of the composition.
- Eclipse: behind and slightly to the right of the monolith.
- Golden particle dunes: occupy the bottom 40% of the composition.
- Human silhouette: in the lower-right-middle area.
- CTA buttons: directly below the hero description.
- Small interactive preview: bottom-right.

Preserve the visual balance between the dark left side and illuminated right side.
Do not stretch the image, crop out the monolith or move the headline into the center.
Reconstruct this composition using actual website components and a responsive Three.js background. The reference image must not simply be used as the entire website background.

FINAL IMPLEMENTATION NOTES

One continuous scene

The desktop homepage should use one persistent Three.js canvas behind the HTML chapters. Do not recreate the entire 3D scene at every section.

Keep the approved hero intact

The reference composition is the visual benchmark. If generated output drifts, use Prompt 10 and the supplied image to bring it back.

Do not fake interactions

A particle effect is only complete when the particles actually respond to pointer and scroll input. A static image is a fallback, not the final interaction.

Protect readability and performance

The visual scene must never block navigation, text selection, buttons or forms. Reduce effects on mobile and respect reduced-motion settings.

Keep content truthful

Use actual screenshots and verified project details. Leave unknown URLs and contact details as editable placeholders.

ARRAKIS

IDEAS. CODE. EXPERIENCES.